# Aditya (Adi) Chinchure

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PhD Candidate in Computer Vision and NLP at UBC, building reasoning-inspired methods to **evaluate and improve vision-language and generative AI models**. My focus spans fine-grained image and video understanding & evaluation, surprise localization and error detection in videos, and improving the diversity and quality of image and video generation models.

## EDUCATION

**University of British Columbia** — *Vancouver, BC*
**PhD in Computer Science** — May 2024 – Present
*Supervisors: Dr. Leonid Sigal & Dr. Vered Shwartz*
◇ Proposed Thesis: Reasoning-Inspired Methods for Evaluating and Improving Generative AI

**MSc, Computer Science**, GPA: 4.0 (A+) — 2024
*Supervisors: Dr. Leonid Sigal & Dr. Renjie Liao*
◇ Thesis: Visual Question Answering with Contextualized Commonsense Knowledge

**BSc, Computer Science, Honors**, GPA: 3.9 (A) — 2021
◇ Thesis: Refinement Architectures for Referring Image Segmentation

## TECHNICAL SKILLS

**Methods** VLM/LLM post-training (SFT, RLHF, GRPO), diffusion and flow-matching models, reward modeling, automatic & human evaluation, distributed multi-GPU training (DeepSpeed)
**Stack** Python, PyTorch, HuggingFace (transformers, diffusers, TRL), vLLM, NumPy/pandas/sklearn, W&B, Slurm

## RELEVANT WORK EXPERIENCE

**UBC Computer Vision & NLP Lab** — May 2024 – Present
*Graduate Research Assistant* | *Supervisor: Dr. Leonid Sigal, Dr. Vered Shwartz* — *Vancouver, BC*
Current research: (a) **distilling fine-grained video understanding** and tool-use capabilities in VLMs, (b) localized-reward **GRPO for video generation** (e.g. LTX-2), (c) designing **VLM-based deepfake video detection** methods on a CIFAR project for AI safety.

**Ideogram** — Nov 2025 – Mar 2026
*Machine Learning Research Intern* | *Supervisor: Dr. Mohammad Norouzi* — *Toronto, ON*
Core research team member for **Ideogram 4.0**, ranked *#1 open-weight model on DesignArena*, with native layout specification. Designed 3+ VLM-based metrics and agentic pipelines for fine-grained text-to-image evaluation, RL, and system prompt optimization.

**Toyota Technological Institute at Chicago** — Oct 2024 – Dec 2024
*Visiting Student Researcher* | *Supervisor: Dr. Matthew Turk* — *Chicago, IL, USA*
Built **InterMit**, a counterfactual method for intersectional bias mitigation in image generation: lower bias (0.33 vs. 0.52; higher diversity), in fewer steps (2.38 vs. 3.15), with better image quality than prior work.

**Borealis AI, Royal Bank of Canada** — Sep 2022 – May 2023
*Machine Learning Research Intern* | *Supervisors: Dr. Fred Tung, Dr. Leonid Sigal* — *Vancouver, BC*
Developed **PD-EST**, a process-disentangling transformer for event sequences such as financial transactions. Jointly learns a process mask and the sequence model, improving next-event time/type prediction with interpretable process decomposition.

**Hypercontext** — May 2019 – Aug 2019
*ML Engineer (Co-op)* — *Toronto, ON*
Developed BERT models for text classification, sentiment analysis and entity recognition. Orchestrated an end-to-end pipeline including data cleanup, a Flask REST API for serving, and retraining and deployment with Docker on AWS.

## PUBLICATIONS

**Spotlight: Identifying and Localizing Video Generation Errors Using VLMs**
**Aditya Chinchure**, Sahithya Ravi, Pushkar Shukla, Vered Shwartz, Leonid Sigal
*European Conference on Computer Vision,* **ECCV 2026** | Paper | Project
◇ 1,600+ annotated video generation errors, including temporal localization, on videos from Veo 3, LTX-2 and Seedance (700+ downloads on Hugging Face in 2 months). VLM-based, inference-time, and multi-agent strategies for error detection and localization.

**SPIKE-RL: Video-LLMs meet Bayesian Surprise**
Sahithya Ravi, **Aditya Chinchure**, Raymond Ng, Leonid Sigal, Vered Shwartz
*International Conference on Learning Representations,* **ICLR 2026** | Paper | Project
◇ Bayesian-surprise frame sampling and GRPO-based belief optimization for Video-LLMs; improved performance over uniform sampling on five downstream benchmarks.

**Mitigate One, Skew Another? Tackling Intersectional Biases in Text-to-Image Models**
Pushkar Shukla*, **Aditya Chinchure***, Emily Diana, Alexander Tolbert, Kartik Hosanagar, Vineeth N Balasubramanian, Leonid Sigal, Matthew A. Turk
*Empirical Methods in Natural Language Processing, Findings,* **EMNLP 2025** | Paper
◇ Introduces BiasConnect, the first intersectional bias analysis in T2I models using concept-level counterfactuals, and InterMit, which uses intersectionality to achieve stronger bias mitigation with fewer steps.

**Black Swan: Abductive and Defeasible Video Reasoning in Unpredictable Events**
**Aditya Chinchure***, Sahithya Ravi*, Raymond Ng, Vered Shwartz, Boyang Li, Leonid Sigal
*Computer Vision and Pattern Recognition,* **CVPR 2025** | Paper | Project & Leaderboard
◇ Video reasoning benchmark spanning 1,655 videos and 15,400+ questions testing explanation of unexpected events and belief revision. Led to the Black Swan Challenge @ CogVL Workshop, CVPR 2026.

**From Local Concepts to Universals: Evaluating the Multicultural Understanding of Vision-Language Models**
Mehar Bhatia, Sahithya Ravi*, **Aditya Chinchure***, Eunjeong Hwang, Vered Shwartz
*Empirical Methods in Natural Language Processing,* **EMNLP 2024** | Paper | Project
◇ Evaluating multicultural understanding in VLMs: cross-cultural image retrieval across 50 countries and culture-specific visual grounding across 15, exposing wide performance gaps and cultural bias.

**TIBET: Identifying and Evaluating Biases in Text-to-Image Generative Models**
**Aditya Chinchure***, Pushkar Shukla*, Gaurav Bhatt, Kiri Salij, Kartik Hosanagar, Leonid Sigal, Matthew A. Turk
*European Conference on Computer Vision,* **ECCV 2024** | Paper | Project
◇ Automatic, prompt-specific bias quantification for image generation using counterfactual reasoning. Works with any black-box T2I model, and achieves concept-level explainability.

**VLC-BERT: Visual Question Answering with Contextualized Commonsense Knowledge**
Sahithya Ravi*, **Aditya Chinchure***, Leonid Sigal, Renjie Liao, Vered Shwartz
*Winter Conference on Applications of Computer Vision,* **WACV 2023** | Paper | Code
◇ Vision-language-commonsense transformer that generates, selects and encodes question-specific contextual knowledge from COMET, outperforming static knowledge-base models on OK-VQA and A-OKVQA.

** indicates equal contribution*

## OTHER PUBLICATIONS

**Position: World Models must live in Parallel Worlds.** Sahithya Ravi*, **Aditya Chinchure***, Pushkar Shukla, Vered Shwartz, Leonid Sigal. *NeurIPS Workshop (LAW),* **NeurIPS-W 2025** | Paper
**PD-EST: Process-disentangling Event Sequence Transformer. Aditya Chinchure**, Fred Tung, Leonid Sigal. *Borealis AI Report*
**LEAP: Private and Federated Data Analysis for Healthcare.** Matheus Stolet, Chris Yoon, Kalli Leung, **Aditya Chinchure**, Mathias Lécuyer, Aline Talhouk, Ivan Beschastnikh. *Poster, BC's AI Showcase, UBC CAIDA* | Project

## AWARDS, SCHOLARSHIPS, AND GRANTS

**CIFAR AI Safety Solution Network on Synthetic AI, Research Grant**, UofT — Sep 2026 – Present
Grant for developing methods for deepfake detection in videos. $12,654 per year.

**Four Year Doctoral Fellowship**, UBC — May 2024 – Present
For outstanding students joining UBC for their doctoral studies. $24,000 per year + tuition.

**Vector Institute Research Grant** — 2022 – Present
Yearly research grant ($6,000) for graduate students affiliated with the Vector Institute for AI.

**Nominated for CRA Outstanding Undergraduate Researcher Award** — Oct 2020
Nominated by Dr. Ivan Beschastnikh for my work on the LEAP project.

**Faculty of Science – International Student Scholarship**, UBC — Sep 2019
A monetary award of $10,000 for strong academic achievement in computer science.

## SERVICE, TEACHING, AND OTHER ACTIVITIES

**Workshops:** Lead organizer, CogVL at CVPR 2026, ft. Trevor Darrell, Katerina Fragkiadaki, Judy Fan & Alane Suhr.

**Mentorship & Talks:** Mentoring three undergrad and master's students on multimodal projects. Talks at TU Delft and CIFAR AI Safety Solution Network on Synthetic AI (2026); guest lectures in CPSC 532V: Commonsense Reasoning (2025–26).

**Teaching Assistant:** Computer Vision (CPSC 425), Intro to AI (CPSC 322), and three database courses at UBC.

**Reviewing:** CVPR*, ECCV*, ICCV, NeurIPS, AAAI, TPAMI and workshops at CVPR and ICCV (2022–2026). **Outstanding reviewer designation at ECCV 2024 and CVPR 2025.*

**Photography:** 260M+ views on Unsplash; professional commercial work. Informs my research on image & video generation.